

STEVENSVILLE, MD, USA

WMO: 720384

Lat: 38.967N

Lon: 76.333W

Elev: 5

StdP: 101.26

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 00124

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.8	(c) -6.1	(d) -18.0	(e) 0.8	(f) -6.6	(g) -15.9	(h) 0.9	(i) -5.2	(j) 11.7	(k) -1.4	(l) 10.8	(m) 0.0	(n) 4.9	(o) 300	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.9	(c) 32.8	(d) 24.2	(e) 32.0	(f) 23.9	(g) 30.3	(h) 23.1	(i) 25.7	(j) 30.2	(k) 25.1	(l) 29.3	(m) 24.5	(n) 28.6	(o) 3.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.8	19.9	27.8	23.8	18.7	27.2	22.9	17.6	26.5	79.5	30.0	76.3	29.0	73.7	28.5	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	8.1	7.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-11.4	27.0	2.3	1.2	-13.0	27.8	-14.3	28.5	-15.6	29.1	-17.3	30.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	2.0	3.4	7.1	13.6	19.0	23.7	26.0	24.8	21.8	15.8	9.0	5.0	(6)
(7)		DBStd	9.31	4.69	4.91	4.89	4.45	3.91	2.88	2.43	2.40	3.25	4.31	4.05	4.25	(7)
(8)		HDD10.0	811	251	193	117	15	0	0	0	0	0	4	65	166	(8)
(9)		HDD18.3	2273	507	419	347	152	41	2	0	0	7	103	281	414	(9)
(10)		CDD10.0	2384	2	7	28	124	278	410	495	458	354	184	34	9	(10)
(11)		CDD18.3	804	0	0	0	11	61	162	237	199	110	24	1	0	(11)
(12)		CDH23.3	6126	0	1	1	72	371	1196	2206	1524	649	106	1	0	(12)
(13)		CDH26.7	1808	0	0	0	11	86	347	759	428	158	19	0	0	(13)
(14)	Wind	WSAvg	3.0	3.4	3.4	3.4	3.6	2.9	2.7	2.5	2.2	2.5	3.0	3.3	3.2	(14)
(15)	Precipitation	PrecAvg	1168	89	67	108	92	94	107	113	114	109	98	80	94	(15)
(16)		PrecMax	1673	176	195	235	167	209	322	215	207	312	209	166	208	(16)
(17)		PrecMin	896	35	7	0	39	19	56	40	30	14	3	14	35	(17)
(18)		PrecStd	176	34	42	48	31	51	60	50	47	69	62	47	44	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	20.1	22.5	27.8	31.4	33.9	35.2	33.9	32.4	28.7	21.4	17.9	(19)
(20)			MCWB	12.9	14.9	15.0	19.8	23.5	24.1	25.7	24.2	23.6	22.7	16.7	16.1	(20)
(21)		2%	DB	13.0	16.2	19.9	25.8	28.9	32.1	32.9	32.2	30.2	26.2	18.7	16.1	(21)
(22)			MCWB	10.7	12.7	13.9	17.7	22.3	23.5	24.6	23.7	22.8	21.0	15.5	14.4	(22)
(23)		5%	DB	11.2	12.8	17.4	22.7	27.2	30.2	32.1	30.9	28.0	24.0	17.1	13.7	(23)
(24)			MCWB	8.7	9.4	12.5	16.8	20.9	22.8	24.5	23.3	22.0	20.9	13.7	12.1	(24)
(25)		10%	DB	8.0	10.9	15.0	21.1	25.2	28.0	30.9	28.9	27.1	22.4	15.2	11.3	(25)
(26)			MCWB	5.7	7.5	11.0	16.4	19.5	21.9	23.6	22.7	21.3	19.1	11.9	9.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.3	16.3	16.9	21.6	24.5	25.6	27.2	26.5	25.4	24.0	18.3	17.3	(27)
(28)			MCD B	16.1	18.7	19.4	25.6	29.8	31.0	32.2	31.2	29.5	26.7	20.2	17.8	(28)
(29)		2%	WB	12.1	13.5	15.2	19.9	23.0	24.6	26.1	25.5	24.1	22.5	16.3	14.9	(29)
(30)			MCD B	13.3	16.2	18.1	23.3	27.3	29.5	30.9	29.3	27.9	24.9	17.6	15.9	(30)
(31)		5%	WB	8.6	10.2	13.5	18.2	21.9	24.0	25.4	24.6	23.4	21.2	14.7	12.5	(31)
(32)			MCD B	10.7	12.3	16.8	21.8	25.9	28.5	29.7	28.3	26.7	23.5	16.5	13.8	(32)
(33)		10%	WB	6.6	7.7	11.7	16.7	21.0	23.3	24.8	23.9	22.6	19.6	12.8	10.0	(33)
(34)			MCD B	8.2	9.7	14.3	20.1	24.0	27.2	29.0	27.6	25.8	22.0	14.7	10.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.8	7.3	8.2	9.3	8.7	8.2	7.9	7.6	7.7	7.6	7.2	6.4	(35)
(36)			MCD BR	10.8	11.4	11.7	12.0	10.3	9.6	9.2	9.0	8.9	8.9	9.3	8.9	(36)
(37)			MCWBR	9.1	8.5	8.0	6.4	5.1	4.0	3.5	4.0	4.0	4.9	6.8	8.5	(37)
(38)		5% WB	MCD BR	10.3	11.0	11.1	11.1	9.4	8.9	8.2	7.8	7.8	7.8	8.2	8.7	(38)
(39)			MCWBR	9.4	9.4	8.5	7.0	4.8	4.1	3.6	3.8	4.1	5.1	7.2	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.312	0.320	0.361	0.398	0.443	0.494	0.518	0.511	0.422	0.362	0.326	0.311	(40)	
(41)		taud	2.479	2.445	2.385	2.301	2.209	2.111	2.060	2.064	2.288	2.459	2.516	2.524	(41)	
(42)		Ebn at Noon	864	905	896	879	842	797	773	767	821	846	841	840	(42)	
(43)		Edh at Noon	79	93	110	127	142	157	164	160	120	92	76	71	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.09	2.94	3.90	4.98	5.61	6.20	6.11	5.42	4.41	3.29	2.38	1.83	(44)	
(45)		RadStd	0.17	0.33	0.31	0.38	0.67	0.47	0.29	0.31	0.49	0.35	0.20	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (80 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+132	+78

Nomenclature: See separate page